



PROGRAMMING FUNDAMENTALS.

C++ I.

1. Variables and Control Flow

Question:

Write a C++ program that reads an integer from the user and prints whether the number is **positive, negative, or zero**.

2. Functions and Modularity

Question:

Create a function that receives two integers and returns the **larger value**.
How would you call this function from `main()`?

3. Arrays and Loops

Question:

Given an array of integers, write code to find the **sum and average** of its elements.
What loop structure would you use and why?

4. Pointers (Core Concept)

Question:

What is a pointer in C++?

Write a small example where a function uses a pointer to **modify the value of a variable declared in `main()`**.

5. Structs (Basic Data Organization)

Question:

Define a `struct` called `Student` with the fields: `name`, `id`, and `grade`.
Write a program that creates one student and prints its information.

```
/*=====
C++ LEVEL 1 INTERVIEW / ASSESSMENT INSTRUCTIONS
=====*/
```

```
/*
SUBMISSION REQUIREMENTS
-----
```

1. Submit ONLY ONE file named:
 `interview11.cpp`
2. The file must contain:
 - ALL questions
 - ALL answerswritten as MULTILINE COMMENTS.

3. Use the following format for EACH question:

```
/*
QUESTION X:
<write the full question here>

ANSWER:
<write your explanation here>
*/
```

4. After each answer, provide the corresponding C++ code (if required).

5. Your code must follow proper modular design:
 - Include FUNCTION PROTOTYPES at the top of the file
 - Provide FUNCTION DEFINITIONS below

6. IMPORTANT:
 - DO NOT call the functions in `main()`
 - Functions must be COMPLETE and READY TO WORK

- The code must COMPILE without errors

7. You may include a main() function, but it must remain EMPTY or minimal:

```
int main() {  
    return 0;  
}
```

8. Follow clean coding practices:

- Meaningful variable names
- Proper indentation
- Clear comments where needed

9. Topics covered:

- Variables and control flow
- Functions
- Arrays
- Pointers
- Structs

10. Any violation of submission format may result in grade reduction.

11. The file must compile.

```
/*=====
```

EXPECTED STRUCTURE EXAMPLE (DO NOT COPY, JUST FOLLOW FORMAT)

```
=====*/
```

```
/*
```

QUESTION 1:

Write a function that returns the square of a number.

ANSWER:

This function takes an integer and returns its square by multiplying the number by itself.

```
*/
```

```
int square(int x); // prototype
```

```
int square(int x) { // definition  
    return x * x;  
}
```

/*=====

END OF INSTRUCTIONS

=====*/

Any question? Just email me.
LDV.